Chapter 3, Problem 1RQ

Problem

At node 1 in the circuit of Fig. 3.46, applying KCL gives:

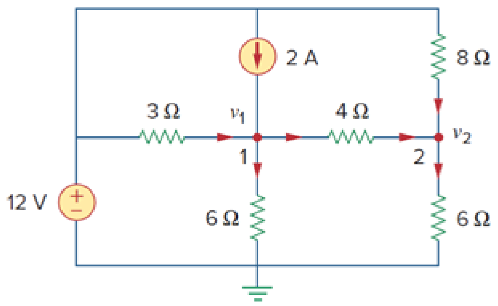
(a) $2+\frac{12-v_1}{3}=\frac{v_1}{6}+\frac{v_1-v_2}{4}$

(b) $2+\frac{v_1-12}{3}=\frac{v_1}{6}+\frac{v_2-v_1}{4}$

(c) $2+\frac{12-v_1}{3}=\frac{0-v_1}{6}+\frac{v_1-v_2}{4}$

(d) $2+\frac{v_1-12}{3}=\frac{0-v_1}{6}+\frac{v_2-v_1}{4}$

Figure 3.46:



Step-by-step solution

Step 1 of 2

Refer to Figure 3.46 in the textbook.

Assume that the current leaving the node leads positive current and the current entering the node leads negative current.

As per Kirchhoff's current law, the sum of the currents entering and leaving the nodes is equal to zero.

Step 2 of 2

Apply Kirchhoff's current law at the node, v_1.

$$-\frac{12-v_1}{3}+\frac{v_1}{6}+\frac{v_1-v_2}{4}-2=0$$

Rearrange the equation.

$$2+\frac{12-v_1}{3}=\frac{v_1}{6}+\frac{v_1-v_2}{4}$$

Thus, the correct answer is $\boxed{(\mathrm{a})}$.